Dear Editors:

We would like to re-submit the revised manuscript, previously reviewed under TVCG-2026-04-0389, as a new submission for consideration to be published on ***IEEE Transactions on Visualization and Computer Graphics***. Following the previous TVCG reviews, we have carefully revised the manuscript in terms of mathematical clarity, methodological justification, experimental settings, efficiency discussion, and ablation analysis. The further information about the paper is in the following:

The Title: GS-LSS: Learned Stochastic Sampling for 3D Gaussian Splatting under Biased Viewpoints

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This manuscript is a study on a practical and underexplored limitation of 3D Gaussian Splatting (3DGS): the quality degradation of synthesized novel views under biased viewpoints, including viewing angle disparities, discontinuous camera trajectories, and systematic coverage gaps. Given TVCG's strong insteret in theories, algorithms, image-based rendering technologies, and volumetric representation, we believe this manuscript is well aligned with the journal's topic.

Biased viewpoints are common issues in real-world data acquisition due to limited hardware accessibility, physical constraints and environmental factors. Under such conditions, existing 3DGS methods tend to overfit to the observed views, resulting in severe perspective overfitting. To address this issue, we propose GS-LSS, an observed-support regularized learned stochastic optimization framework that reformulates 3DGS training as a Learned Stochastic Sampling (LSS) process. The main contributions of this work are summarized as follows:

- We investigate the behavior of representative 3DGS meth ods under various biased viewpoint scenarios, showing that perspective overfitting is a recurring issue caused by the under-determination of the per-frame optimization in weakly observed regions..
- We propose GS-LSS, an observed-support regularized LSS framework for 3DGS that mitigates perspective overfitting by complementing view-dependent supervi sion with per-primitive deterministic drift and adaptive stochastic perturbation.
- We propose a Continuous Probability Field (CPF) that serves as a data-adaptive consistency prior, approximating aggregated observed support as a smooth and continuous field for Gaussian relocation and reshaping.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the submission of this manuscript and agree with its contents.

Thank you for your time and consideration. We appreciate your evaluation of our work and look forward to your response.

Sincerely yours,

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